

RETIE, Belgium

WMO: 064640

Lat: 51.2214N

Lon: 5.0272E

Elev: 69

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.9	(c) 24.3	(d) 15.0	(e) 11.8	(f) 22.2	(g) 20.2	(h) 15.2	(i) 26.6	(j) 21.9	(k) 52.2	(l) 19.8	(m) 50.1	(n) 4.6	(o) 70	(p) 0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.2	(c) 86.4	(d) 70.6	(e) 81.7	(f) 68.0	(g) 78.5	(h) 66.3	(i) 72.9	(j) 82.1	(k) 70.3	(l) 78.2	(m) 68.2	(n) 75.8	(o) 5.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	110.5	76.9	67.6	101.8	73.9	65.6	94.8	71.3	36.6	81.9	34.3	77.8	32.5	76.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a) 17.0	(b) 14.6	(c) 12.9	(d) DB	(e) 14.1	(f) 94.5	(g) 7.9	(h) 3.8	(i) 8.5	(j) 97.2	(k) 3.8	(l) 99.5	(m) -0.6	(n) 101.6	(o) -6.3	(p) 104.4	
(4) 17.0	(5) 14.6	(6) 12.9	(7) WB	(8) 13.8	(9) 76.3	(10) 7.8	(11) 3.4	(12) 8.2	(13) 78.8	(14) 3.6	(15) 80.7	(16) -0.8	(17) 82.6	(18) -6.5	(19) 85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	38.4	39.0	44.3	50.5	56.3	62.4	65.9	64.3	59.1	53.7	45.1	40.1	(6)
	DBStd	11.59	7.27	7.60	6.54	6.74	6.83	5.55	5.91	4.89	4.98	6.38	6.12	7.08	(7)
	HDD50	1473	361	309	193	77	15	0	0	0	1	36	170	311	(8)
	HDD65	5066	824	728	640	436	282	115	59	72	189	351	597	773	(9)
	CDD50	2083	3	1	18	91	212	373	492	442	273	151	23	4	(10)
	CDD65	201	0	0	0	1	14	38	86	49	11	2	0	0	(11)
	CDH74	2052	0	0	0	32	142	393	870	486	114	15	0	0	(12)
	CDH80	592	0	0	0	1	20	98	294	154	25	0	0	0	(13)
(14) Wind	WSAvg	5.7	7.1	7.2	6.6	5.8	5.2	4.8	4.4	4.4	4.4	5.3	6.1	7.3	(14)
(15) Precipitation	PrecAvg	31.9	2.7	2.6	2.0	1.8	2.4	2.9	3.1	3.1	2.4	2.6	3.0	3.2	(15)
	PrecMax	40.7	5.6	5.9	4.2	3.4	4.5	8.5	5.9	6.8	6.7	7.2	5.1	6.1	(16)
	PrecMin	24.1	0.1	0.5	0.1	0.1	0.8	0.6	1.1	1.0	0.5	0.9	0.6	1.1	(17)
	PrecStd	3.9	1.3	1.3	1.1	0.8	1.0	1.5	1.4	1.8	1.6	1.2	1.1	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	60.0	68.0	77.7	82.4	88.2	93.6	90.8	83.1	76.2	63.1	56.4	(19)
		MCWB	54.1	48.8	55.6	60.3	66.8	70.7	72.6	71.8	69.5	64.1	58.6	53.2	(20)
	2%	DB	53.5	53.1	63.3	73.4	77.9	82.2	87.7	84.2	76.7	70.6	59.1	54.3	(21)
		MCWB	51.8	48.9	53.8	58.8	63.7	67.8	71.6	69.7	66.1	63.8	56.8	52.4	(22)
	5%	DB	50.7	51.3	58.8	68.3	73.9	78.5	82.1	79.1	72.5	65.9	56.6	51.6	(23)
		MCWB	49.1	48.5	52.0	57.5	61.9	66.4	68.7	67.6	64.6	60.9	54.5	50.1	(24)
	10%	DB	48.4	49.3	54.7	62.9	69.6	74.4	78.6	74.9	68.6	62.7	54.2	49.3	(25)
		MCWB	47.2	47.1	49.6	54.7	59.9	64.3	66.8	65.2	62.0	59.4	52.4	47.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	52.0	57.1	63.0	70.1	74.5	75.7	75.3	72.2	67.4	59.6	54.8	(27)
		MCDB	55.5	54.3	64.5	73.0	78.8	84.2	86.6	84.2	79.7	73.6	61.5	55.5	(28)
	2%	WB	51.9	50.2	55.1	60.5	66.0	70.8	73.3	71.6	68.1	63.9	57.3	52.5	(29)
		MCDB	53.0	52.4	62.0	70.1	75.0	78.2	83.7	79.9	74.3	68.3	59.0	53.9	(30)
	5%	WB	49.3	48.9	52.7	58.0	63.3	67.7	70.8	68.9	65.6	61.8	55.0	50.4	(31)
		MCDB	50.4	51.1	57.7	66.2	70.9	75.4	79.5	76.1	70.5	64.8	56.3	51.6	(32)
	10%	WB	47.2	47.0	50.5	55.7	60.9	65.5	68.3	66.6	63.4	60.1	52.5	48.3	(33)
		MCDB	48.3	49.3	54.1	62.7	67.7	72.8	76.4	72.7	67.6	62.7	53.9	49.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	11.4	15.5	18.8	18.1	18.1	19.2	18.5	17.6	13.7	10.3	8.6	(35)
		MCDBR	9.8	13.8	23.5	26.8	24.7	25.4	27.1	26.5	23.5	18.1	11.5	9.6	(36)
		MCWBR	8.7	10.3	15.0	14.3	12.5	12.4	12.3	13.2	13.9	11.6	9.5	8.5	(37)
	5% WB	MCDBR	9.7	12.6	20.7	23.7	21.3	21.8	23.9	22.5	20.7	15.9	11.2	9.1	(38)
		MCWBR	8.9	9.9	14.0	13.5	12.4	12.1	12.1	13.0	13.7	11.1	9.7	8.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.334	0.358	0.392	0.422	0.431	0.428	0.435	0.425	0.397	0.371	0.347	0.332	(40)
	taud		2.358	2.320	2.229	2.179	2.201	2.229	2.214	2.262	2.329	2.380	2.367	2.329	(41)
	Ebn at Noon		213	239	252	258	262	263	259	255	249	232	205	192	(42)
	Edh at Noon		21	27	36	42	43	43	42	38	32	26	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		240	455	824	1318	1578	1711	1610	1374	1021	601	282	186	(44)
	RadStd		31	89	117	145	138	142	193	128	98	48	30	25	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page